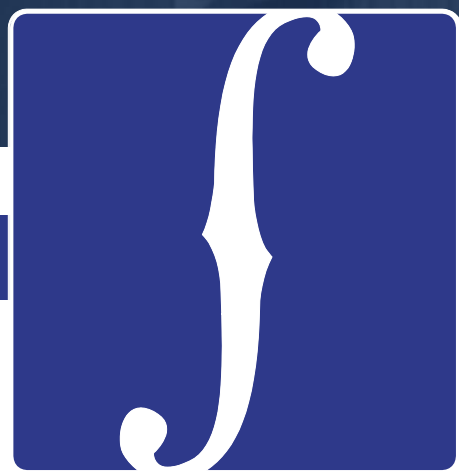




CELLO

CELLO RESEARCH

DECEMBER 2015

- I. The Case for In-House Research
- II. Areas of Focus
- III. The Role of Technology
- IV. Team Bios

THE CASE FOR IN-HOUSE RESEARCH

- **Alpha Generation:** Generate original, consensus-defying ideas through researching homeowner prepayment behavior, MBS pricing models, and industry trends that affect MBS pricing
- **Accuracy:** Street research can oversimplify the complexities of prepayment data. “Paper” trades often do not accurately capture the dynamics of pricing and hedging an MBS trade from inception to close. The Cello Research team serves as a valuable resource in cross-validating/fact-checking Street research.
- **Speed:** Proprietary databases and pricing models help us sift through large datasets very rapidly to analyze the idiosyncratic risk associated with every CMO. Research analysts leverage technology to quickly address queries from the investment team.

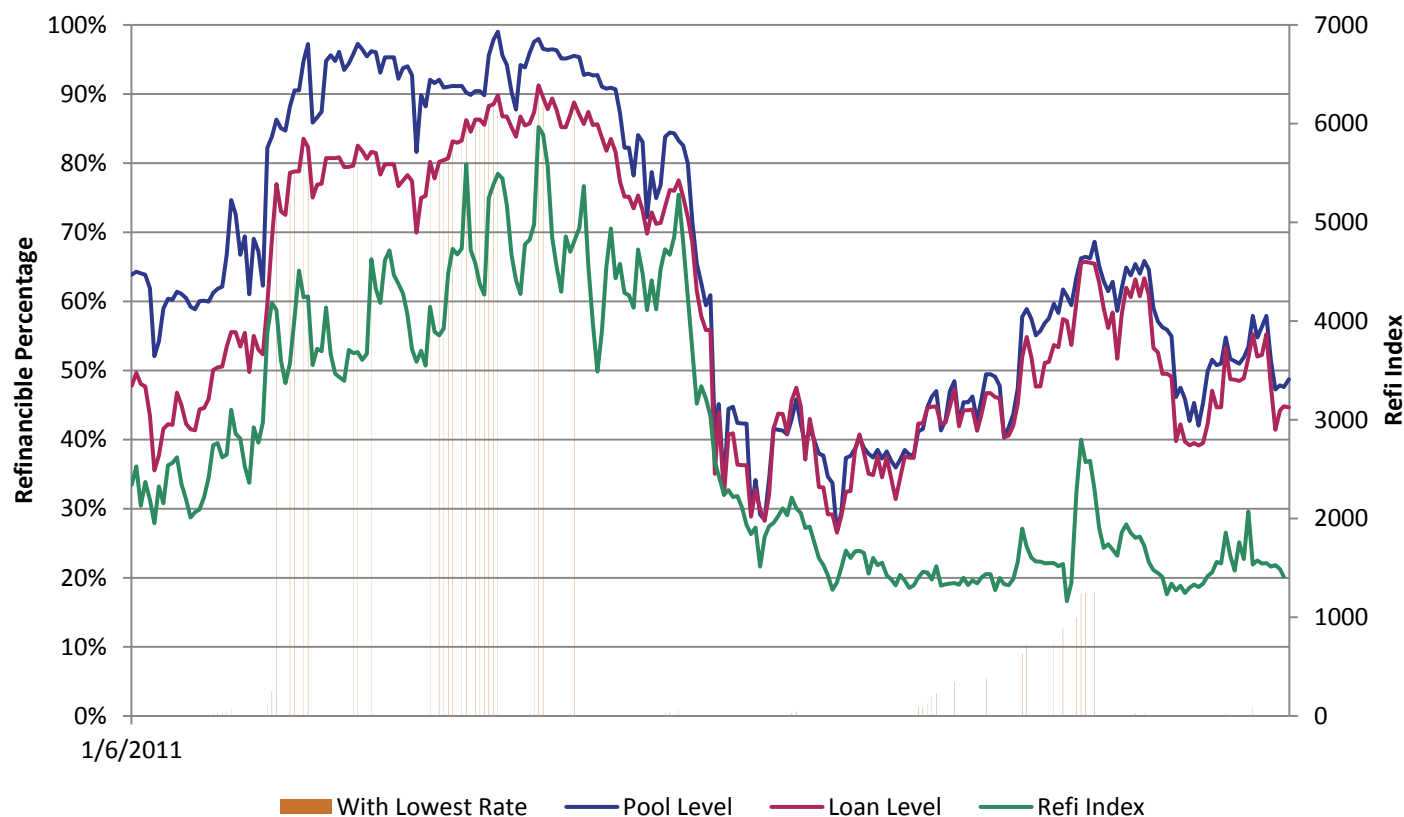
AREAS OF FOCUS

- **Prepayment Behavior:** Evolution of Borrower/Loan characteristics, Government housing policy, Mortgage lending industry, Prepayment modeling
- **MBS CMO Valuation and Hedging:** Mortgage pricing models, Interest rate models, Mortgage rate models, House price models
- **Relative Value Models:** Econometric modeling of price/spread relationships for various fixed-income securities
- **Scale (with IT):** Software framework for pricing, hedging and risk management of fixed-income instruments and, in particular, structured products arising in the housing markets

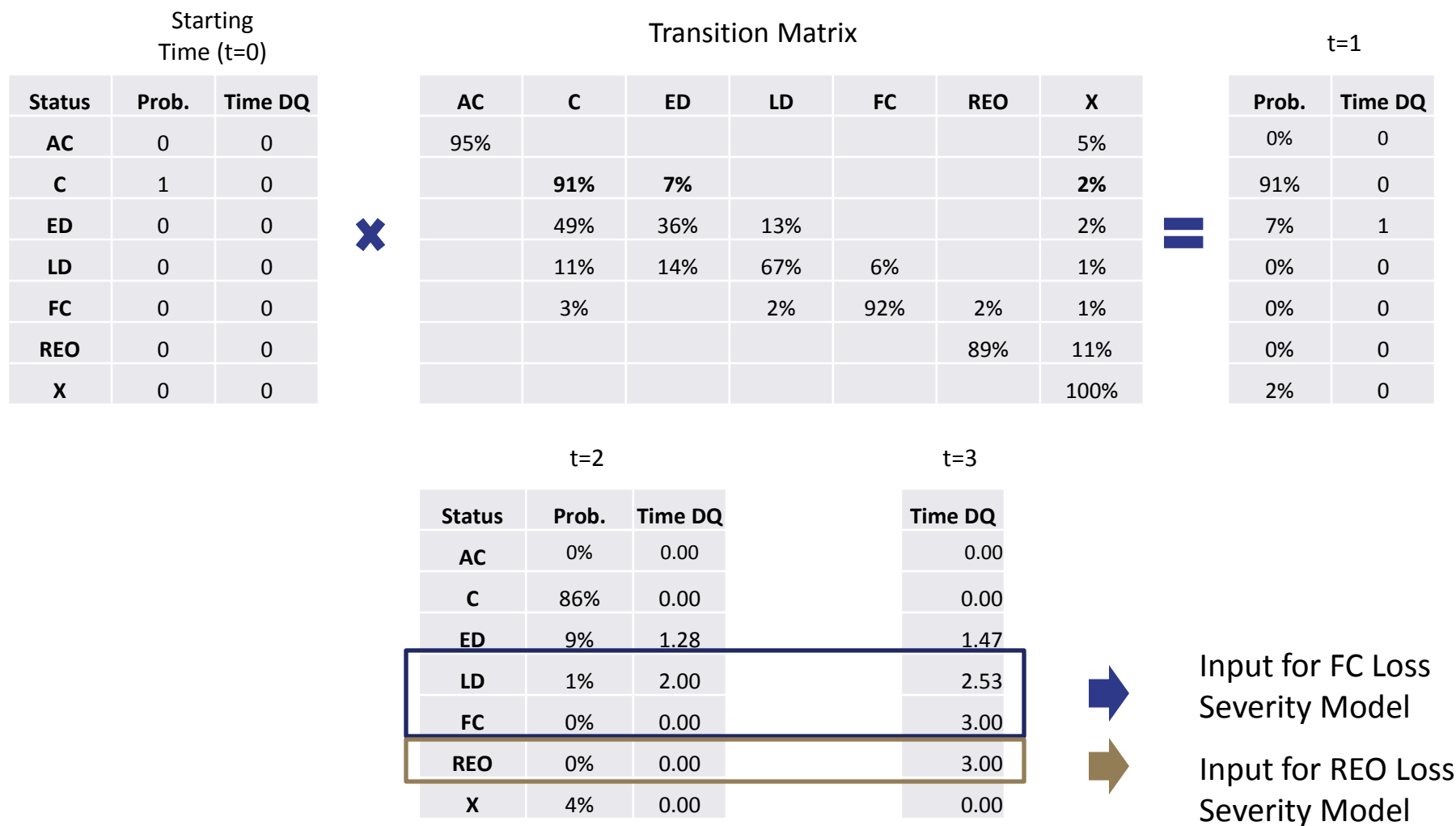
The number of borrowers who are eligible to refinance at prevailing mortgage rates and underwriting standards is a critical determinant of the level of prepayment risk in the market. Once a critical threshold of borrowers is eligible, there is often a surge in prepayment rates due to media coverage of refinancing opportunities and solicitation of refinances by originators.

Cello calculates the Refinanceable Universe in a number of different ways. In particular, the loan-level Refinanceable Universe calculation explicitly takes into account loan-level pricing adjustments and PMI-related costs.

THE REFINANCEABLE UNIVERSE VS THE REFINANCE INDEX (January 2011 – December 2015)



Stylized Single Loan Calculation

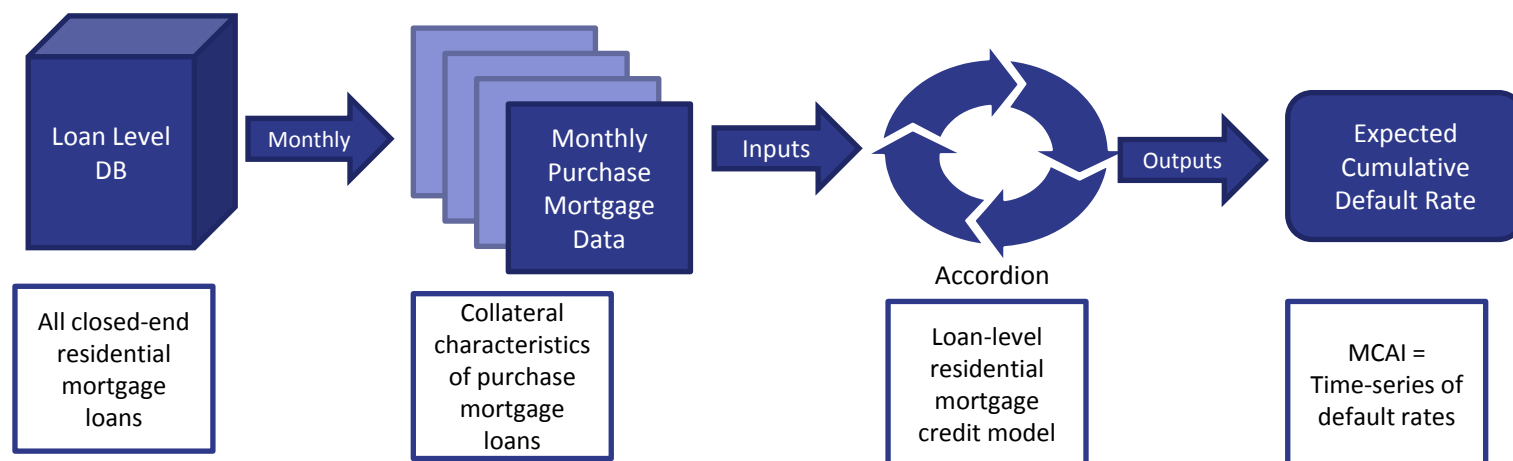


Cello's Mortgage Credit Availability Index (MCAI) assumes that the market increases/decreases credit availability by taking on more/less mortgage default risk.

The fundamental advantage of following this method is that it provides a single comprehensive measure at any point in time that aggregates across the different risk factors that are taken into account in the typical underwriting process (FICO, LTV, DTI, etc.).

The expected default rate is a weighted average of the defaults projected over a "normal" and "stressed" environment.

CONSTRUCTING CELLO'S MORTGAGE CREDIT AVAILABILITY INDEX

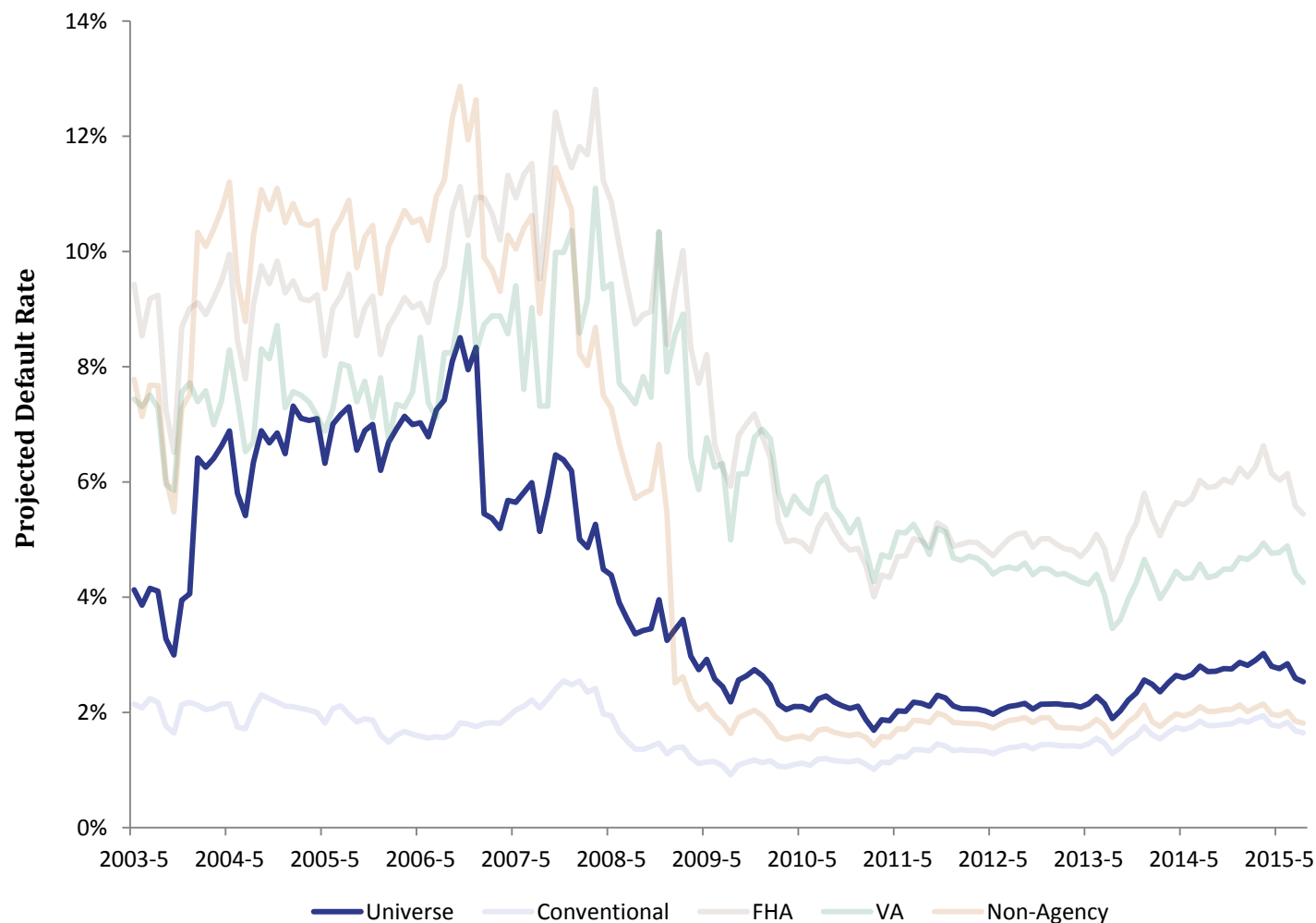


Cello's MCAI shows that, on an aggregate basis, mortgage credit availability at its 2007 peak was almost 4x its prevailing levels.

Most of the post-2007 decline in MCAI has been driven by a substantial decrease in Product Risk with a significant slump in the volume of originated Alt-A and Subprime loans.

More recently (starting 2014), there has been a gradual improvement in mortgage credit availability but overall levels are considerably lower than in the past.

CELLO'S MORTGAGE CREDIT AVAILABILITY INDEX



Cello has developed a quantitative framework for managing IO Beta exposure. The goal is to design a strategy that outperforms a passive strategy that is always 100% invested.

A passive IO Beta Strategy runs the risk of significant drawdowns. Over the period March 2011 to November 2015, Cello estimates that the passive strategy had a worst month of down 12.9% and a cumulative maximum drawdown of 23.5%.

Quantitative Framework for Active IO Beta Strategy

- **Utilize Model-free RV Metrics:** OAS and Prepayment modeling are already heavily used in allocating Beta so the quantitative approach provides a different viewpoint.
- **RV Metrics:** Carry, Realized Volatility, Media Effect, IOS Prices
- **Utilize IOS as IO Benchmark:** The model does not capture the fact that the Cash-Synthetic basis may decouple.
- **Optimize Sharpe Ratio:** Algorithmically select an active management strategy that results in the highest Sharpe Ratio

Current Forecast

- **Model Recommended IO Beta Exposure** (as of December 11th): 0-5%
- **Fund Exposure** (as of December 11th): 18%. The Fund exposure will typically run higher than the model recommendation since Structured IO offers higher OASs than IOS. The Investment Committee can also overlay other factors such as technical conditions and/or a view on macro factors such as interest rates, the slope of the yield curve, the expected level of volatility, the likelihood of a prepayment surprise etc.

- **Scale** is built on top of **Quantlib**, a well maintained open-source C++ library for modeling, trading, and risk management. The approach allows for reducing low-level model building and focusing on complex problems.
- **Scale** allows for micro-level specification of proprietary models.
- **Scale** provides a platform for rapidly assessing new structures and strategies.
- **Scale** serves as a tangible repository for Cello IP.
- **Scale** is a training ground for new Associates/Analysts in the Research group.

THE ROLE OF TECHNOLOGY

Cello has built a comprehensive suite of trading and operations reports based on its robust databases.

Cello continues to make significant investments to build out its proprietary analytic infrastructure. The main highlights includes the following

Symphony Application

- Bond Pricing
- Collateral Analysis(slice and dice)
- Excel API

Scale Models

- Interest Rate Model
- Mortgage Rate Model
- Prepayment Model
- OAS Analytics

Motif Calculation Service

- Distributed Computing
- Amazon Cloud Integration
- In-Memory Caching
- OAS Path Distribution

